WEATHER DATA ANALYSIS

Meteorological Analysis Across Indian States

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1. Executive Summary

This document presents a comprehensive analysis of weather data across 11 Indian states from January 2020 to January 2024. The dataset captures critical meteorological variables — temperature, rainfall, humidity, and wind speed — across four distinct seasons. Key findings are summarised below.

1.1 Dataset

Attribute	Value
Total Records	80 observations
Date Range	Jan 2020 – Jan 2024
States Covered	11 Indian states
Seasons	Winter, Spring, Summer, Autumn
Variables Tracked	13 columns (temp, rainfall, humidity, wind, rainy days)
Missing Values	None — complete dataset

1.2 Key Highlights

- Mean average temperature across all records: 26.1°C (range: 12.0°C – 36.0°C)
- Maximum temperature on record: 42.0°C (Rajasthan, May 2022)
- Highest single rainfall event: 177.4 mm (Kerala, October 2024)
- Summer is the wettest season with an average of 104.6 mm rainfall per observation
- Kerala records the highest rainfall (avg 102.3 mm) and humidity (avg 78.1%)
- Rajasthan is the driest state with only 6.7 mm average rainfall
- No missing values detected across all 13 columns — dataset is complete

2. Project Overview

2.1 Objectives

The primary objectives of this weather data analysis project are:

- To examine and compare temperature patterns across Indian states and seasons
- To identify high-rainfall regions and understand seasonal rainfall distribution
- To analyse the relationship between humidity, rainfall, and rainy days
- To surface outliers and extreme weather events within the dataset
- To provide a reusable Excel workbook for ongoing meteorological monitoring

2.2 Workbook Structure

The Excel file WEATHER__DATA_ANALYSIS.xlsx is organised into 7 purpose-built sheets:

Sheet	Type	Description
MAIN DATA	Raw Data	Primary dataset — 80 rows, 13 columns
DASH BOARD	Visual	Charts and summary KPI indicators
PIVOT TABLE	Aggregation	Monthly and seasonal aggregations with pivot summaries
TEMPERATURE	Analysis	Focused deep-dive into temperature metrics
HUMIDITY	Analysis	Focused deep-dive into humidity metrics
WEATHER	Analysis	Combined multi-variable weather pattern analysis
RAINFALL	Analysis	Focused deep-dive into rainfall metrics

2.3 Data Schema

Each observation in the MAIN DATA sheet captures the following 13 attributes:

Column Name	Data Type	Description
Date	Date	Observation date (DD-MMM-YYYY)
Month	Text	Month name (January – December)
Season	Text	Season: Winter / Spring / Summer / Autumn
State	Text	Indian state of observation
Avg Temp (°C)	Numeric	Mean daily temperature in Celsius
Max Temp (°C)	Numeric	Maximum daily temperature in Celsius
Min Temp (°C)	Numeric	Minimum daily temperature in Celsius
Rainfall (mm)	Numeric	Daily recorded rainfall in millimetres
Humidity (%)	Numeric	Relative humidity as a percentage
Wind Speed (km/h)	Numeric	Wind speed in kilometres per hour
Total Rainfall (mm)	Numeric	Cumulative rainfall for the period
Avg Rainfall (mm)	Numeric	Average rainfall for the period
Rainy Days	Integer	Count of rainy days in the observation period

3. Statistical Analysis

3.1 Descriptive Statistics

The table below summarises descriptive statistics for all numeric variables across the 80 observations:

Metric	Count	Mean	Std Dev	Min	Median	Max
Avg Temp (°C)	80	26.1	5.1	12.0	26.3	36.0
Max Temp (°C)	80	30.9	5.3	16.9	31.0	42.0
Min Temp (°C)	80	21.5	5.3	7.0	21.5	31.0
Rainfall (mm)	80	56.7	46.6	4.0	47.0	177.4
Humidity (%)	80	64.9	13.6	38.0	68.0	92.0
Wind Speed (km/h)	80	18.3	3.1	10.0	18.5	28.0
Rainy Days	80	2.25	1.82	0	2.0	6

3.2 Correlation Analysis

The correlation matrix below reveals the strength of linear relationships between numeric variables. A strong positive correlation (0.78) exists between Humidity and Rainfall, confirming that wetter conditions are consistently accompanied by higher humidity levels.

Variable	Avg Temp	Rainfall	Humidity	Wind Spd	Rainy Days
Avg Temp (°C)	1.00	0.30	0.14	-0.11	-0.01
Rainfall (mm)	0.30	1.00	0.78	0.10	0.53
Humidity (%)	0.14	0.78	1.00	0.12	0.56
Wind Speed (km/h)	-0.11	0.10	0.12	1.00	0.03
Rainy Days	-0.01	0.53	0.56	0.03	1.00

4. Regional Analysis

4.1 Temperature by State

Tamil Nadu records the highest average temperature (29.3°C) while Punjab has the lowest (18.0°C), reflecting strong latitudinal and climatic variation across the subcontinent.

State	Avg Temp (°C)	Avg Rainfall (mm)	Avg Humidity (%)
Tamil Nadu	29.3	62.8	66.1
Odisha	28.4	62.0	68.1
Maharashtra	28.3	51.4	62.7
Rajasthan	26.5	6.7	43.0
Gujarat	26.4	21.3	49.0
Kerala	26.3	102.3	78.1
West Bengal	26.1	56.9	65.1
Karnataka	25.7	47.2	61.7
Uttar Pradesh	24.3	53.2	61.0
Assam	22.5	74.7	73.0
Punjab	18.0	20.3	61.7

4.2 Wind Speed by State

Karnataka leads wind speed averages at 20.4 km/h, while Uttar Pradesh is the calmest state at 15.1 km/h. Coastal and mountainous states tend to experience higher wind exposure.

Karnataka	20.4 km/h
Punjab	19.7 km/h
Assam	19.4 km/h
Rajasthan	18.9 km/h
West Bengal	18.6 km/h
Tamil Nadu	18.4 km/h
Kerala	17.7 km/h
Odisha	17.5 km/h
Maharashtra	17.5 km/h
Gujarat	17.1 km/h

Uttar Pradesh	15.1 km/h

5. Seasonal Analysis

5.1 Season-wise Weather Averages

Summer is the dominant season for both temperature and rainfall. Winter records the lowest temperatures and the least precipitation. The data covers 80 observations, roughly balanced across the four seasons (19–21 records each).

Season	Avg Temp (°C)	Rainfall (mm)	Humidity (%)	Wind Speed (km/h)
Summer	30.5	104.6	75.5	18.8
Spring	28.8	30.4	55.8	18.0
Autumn	25.2	64.4	64.9	17.7
Winter	20.3	24.4	59.7	18.3

5.2 Monthly Temperature Trend

Temperatures peak in the June–August window (~30.5–30.9°C) and dip to their lowest in December–January (~18.8–19.1°C). This follows the expected seasonal arc for the Indian subcontinent.

Month	Avg Temp (°C)	Season
January	18.8	Winter
February	22.9	Winter
March	26.4	Spring
April	30.1	Spring
May	30.4	Spring
June	30.9	Summer
July	29.9	Summer
August	30.6	Summer
September	27.6	Autumn
October	24.5	Autumn
November	23.8	Autumn
December	19.1	Winter

5.3 Seasonal Rainfall Distribution

Summer accounts for 46% of total recorded rainfall (2,092 mm cumulative), making it by far the wettest season. Winter is the driest season with only 512 mm cumulative across 21 observations.

Season	Avg (mm)	Min (mm)	Max (mm)	Cumulative (mm)
Summer	104.6	37.2	177.4 (Kerala)	2,092
Autumn	64.4	12.0	177.4	1,289
Spring	30.4	5.0	70.0	578
Winter	24.4	4.0	62.2	512

6. Extreme Weather Events

6.1 Top 5 Hottest Records

Rajasthan recorded the highest temperature of 42.0°C in May 2022, consistent with its desert climate. Tamil Nadu and Gujarat also appear in the top records during summer months.

#	Date	State	Season	Max Temp	Humidity
1	25-May-2022	Rajasthan	Spring	42.0°C	38%
2	02-May-2023	Tamil Nadu	Spring	40.0°C	60%
3	20-Jun-2025	Gujarat	Summer	39.7°C	50%
4	10-Jun-2020	Tamil Nadu	Summer	39.0°C	80%
5	04-Jun-2026	Kerala	Summer	38.5°C	69%

6.2 Top 5 Highest Rainfall Events

Kerala dominates the highest rainfall records with three out of five top events, all exceeding 150 mm in a single observation. This aligns with Kerala's status as one of India's most rainfall-intensive coastal states.

#	Date	State	Season	Rainfall	Humidity
1	03-Oct-2024	Kerala	Autumn	177.4 mm	88%
2	01-Aug-2021	Assam	Summer	162.0 mm	92%
3	08-Aug-2022	Kerala	Summer	155.0 mm	92%
4	26-Nov-2025	Kerala	Autumn	154.5 mm	79%
5	23-Oct-2026	Tamil Nadu	Autumn	149.5 mm	82%

7. Year-on-Year Trends

7.1 Annual Temperature Trend

Average temperatures remain remarkably stable year-on-year, hovering between 25.5°C and 26.4°C. No strong upward or downward trend is evident within the observed window — likely reflecting the balanced seasonal sampling rather than a long-term climatic signal.

2020	26.3°C
2021	26.3°C
2022	25.8°C
2023	26.2°C
2024	25.9°C
2025	26.4°C
2026	25.5°C

7.2 Annual Rainfall Trend

Average annual rainfall shows a modest declining trend from 2020 to 2024, before rebounding in 2025–2026. Year-to-year variability is driven by differences in which states and seasons are represented in each annual sample.

2020	54.6 mm
2021	54.4 mm
2022	54.2 mm
2023	53.0 mm
2024	50.5 mm
2025	62.5 mm
2026	60.3 mm

8. Key Findings & Recommendations

8.1 Key Findings

- Kerala is the wettest and most humid state — it records the highest average rainfall (102.3 mm), highest humidity (78.1%), and appears in 3 of the top 5 rainfall events.
- Rajasthan is the driest and most extreme state — the driest (6.7 mm avg rainfall) and home to the highest temperature on record (42.0°C).
- Summer drives peak weather extremes — highest temperatures (30.5°C), rainfall (104.6 mm), and humidity (75.5%) all peak in summer.
- Rainfall and humidity are strongly correlated ($r = 0.78$) — humidity is a reliable proxy for expected rainfall.
- Wind speed shows no strong correlation with other weather variables, suggesting it is driven by independent geographic factors.
- The dataset is clean and complete — no missing values across 80 rows and 13 columns.

8.2 Recommendations

- Expand dataset: Increase observations per state and add sub-district or city-level granularity for more precise regional analysis.
- Extend date range: A 10+ year dataset would allow meaningful trend analysis and detection of climate change signals.
- Add forecasting models: Apply time-series models (ARIMA, Prophet) on monthly/seasonal data for rainfall and temperature forecasting.

- Integrate elevation data: Correlating altitude with temperature and rainfall would improve explanatory power across diverse terrain.
- Dashboard enhancement: Add interactive slicers in the Excel dashboard for dynamic filtering by state, season, and year.

9. Appendix

9.1 State Observation Count

The number of observations per state reflects the sampling density in the dataset:

Kerala	10 observations
Assam	9 observations
Tamil Nadu	9 observations
West Bengal	8 observations
Odisha	8 observations
Maharashtra	7 observations
Gujarat	7 observations
Karnataka	7 observations
Rajasthan	6 observations
Uttar Pradesh	6 observations
Punjab	3 observations

9.2 Glossary

Term	Definition
Avg Temp (°C)	The arithmetic mean of all temperature readings for a given observation period
Humidity (%)	Relative humidity — the percentage of water vapour in the air relative to saturation
Rainfall (mm)	Total precipitation measured in millimetres over the observation period
Rainy Days	Count of days within the observation period where measurable precipitation occurred
Wind Speed (km/h)	Average wind speed in kilometres per hour for the observation period
Correlation (r)	Pearson correlation coefficient; values near ±1 indicate strong linear relationships